Multiple IV Infusions & Multiple Oral Practice 2

Module 6 - Multiple Dosings · Daily practice sheet · No date

Name _____ Date _____

1a 1. A 300-mg dose of an antibiotic was administered as an IV infusion to a 85 kg male over a period of 90 minutes. Eight hours after the start of the first infusion, a second 300-mg dose was infused, again over a period of 90 minutes. The drug has a half-life of 3.3 hours and apparent volume of distribution of 0.2 L/kg. a. What is the plasma drug concentration at the end of the first infusion?

1b 1. A 300-mg dose of an antibiotic was administered as an IV infusion to a 85 kg male over a period of 90 minutes. Eight hours after the start of the first infusion, a second 300-mg dose was infused, again over a period of 90 minutes. The drug has a half-life of 3.3 hours and apparent volume of distribution of 0.2 L/kg. b. What is the plasma drug concentration 4 hours after the cessation of the second infusion?

2a 2. An adult male (72 kg) was given 300 mg of an antibiotic orally every 8 hours for 2 weeks. The literature reports that the antibiotic is about 85% bioavailable and has a V_D of 3.2 L/kg. The elimination half-life is about 7.3 hours and the absorption half-life is about 75 minutes. Calculate a. C_{max} after the first dose.

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2c 2. An adult male (72 kg) was given 300 mg of an antibiotic orally every 8 hours for 2 weeks. The literature reports that the antibiotic is about 85% bioavailable and has a V_D of 3.2 L/kg. The elimination half-life is about 7.3 hours and the absorption half-life is about 75 minutes. Calculate c. Maximum plasma drug concentration at steady state.

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Solutions

Her key: the answer and working on her own solved sheet or annotated slide. Worked by the drill: the drill's own working, where her sheet gives no worked value.

1a

Her key

Answer: 15.14 mg/L

WHAT THE STEM GIVES

D_0	300-mg	$R = \frac{300}{1.5} = 200$ mg/hr
weight	85 kg	$V_D = 0.2 \times 85 = 17$ L
infusion time	90 minutes	1.5 hr: the denominator of R and the t in $(1 - e^{-kt})$
second start	Eight hours after the start	*not needed: only the first infusion is asked*
t½	3.3 hours	gives $k = \frac{0.693}{3.3} = 0.21$ hr^{-1}
V_D	0.2 L/kg	times 85 kg gives 17 L; $Cl_T = V_D \times k$

THE WORKING, ONE LINE AT A TIME

UNIT

$$90 \text{ min} = 1.5 \text{ hr}; \; R = \frac{300 \text{ mg}}{1.5 \text{ hr}} = 200 \text{ mg/hr}; \; V_D = (0.2 \text{ L/kg})(85 \text{ kg}) = 17 \text{ L}; \; k = \frac{0.693}{3.3 \text{ hr}} = 0.21 \text{ hr}^{-1}$$

The infusion time is put in hours so that R is in mg/hr and matches k in hr^{-1}; the volume is 0.2 L per kilogram times the weight.

SETUP

$$C_p = \frac{R}{V_D \, k} (1 - e^{-kt}), \text{ with } t = 1.5 \text{ hr}$$

$V_D \times k$ is the clearance, so R over it is the plateau a continuous infusion would reach; $(1 - e^{-kt})$ is the fraction of it reached when this infusion stops.

ALGEBRA

$$C_p = \frac{200 \text{ mg/hr}}{(17 \text{ L})(0.21 \text{ hr}^{-1})} (1 - e^{-(0.21)(1.5)}) = (56.02 \text{ mg/L})(0.2702) = 15.14 \text{ mg/L}$$

mg/hr over L/hr leaves mg/L; 1.5 hours is less than half a half-life, so only 27% of the 56.02 mg/L plateau is reached.

> **DOES THE NUMBER MAKE SENSE?**
> The plateau R over V_D k is 200 over 17 × 0.21, about 56.02 mg/L; 1.5 hours is under half a half-life of 3.3 hours, so well under half of the plateau is reached: 15.14 mg/L fits and lies below 56.02.

1b

Her key

Answer: 7.75 mg/L

WHAT THE STEM GIVES

D_0	300-mg	each infusion ends at 15.14 mg/L, as in part a
weight	85 kg	$V_D = 0.2 \times 85 = 17$ L
infusion time	90 minutes	1.5 hr: sets the end times, 1.5 and 9.5 hr
second start	Eight hours after the start	the second infusion runs from 8 to 9.5 hr
t½	3.3 hours	gives $k = \frac{0.693}{3.3} = 0.21\ \text{hr}^{-1}$
V_D	0.2 L/kg	with k gave the end-of-infusion 15.14 mg/L in part a
t	4 hours after the cessation	the second has declined 4 hr, the first 12 hr

THE WORKING, ONE LINE AT A TIME

SETUP

Time line: 0, 1.5, 8, 9.5, 13.5 hr

The first infusion runs from 0 to 1.5 hours; the second starts at 8 and ends at 9.5 hours; 4 hours after its end is 13.5 hours. The first infusion has then been declining for 12 hours and the second for 4.

SETUP

$C_p = (15.14\ \text{mg/L})\ e^{-(0.21)(12)} + (15.14\ \text{mg/L})\ e^{-(0.21)(4)}$

Each infusion contributes its end-of-infusion concentration declined for the time since its own end; the two add because elimination is first order.

ALGEBRA

$C_p = (15.14)(0.0805) + (15.14)(0.4317) = 1.2188 + 6.5359 = 7.75\ \text{mg/L}$

Twelve hours is 3.6 half-lives, so 8% of the first infusion is left; four hours is 1.2 half-lives, so 43% of the second is left and it supplies most of the sum.

> **DOES THE NUMBER MAKE SENSE?**
> Four hours is a little over one half-life of 3.3 hours, so a little under half of the second infusion remains, 6.5359 mg/L; twelve hours is over three half-lives, leaving about an eighth of the first, 1.2188; the sum 7.75 mg/L lies below 15.14.

2a

Her key

Answer: 0.7687 mg/L

WHAT THE STEM GIVES

weight	72 kg	$V_D = 3.2 \times 72 = 230.4$ L
D_0	300 mg	the dose in the numerator
τ	every 8 hours	*not needed for the first-dose peak*
duration	2 weeks	*not needed: long enough for steady state*
F	85% bioavailable	F = 0.85 in the numerator

V_D	3.2 L/kg	times 72 kg gives 230.4 L, the denominator
t½	7.3 hours	gives $k = \frac{0.693}{7.3} = 0.0949\ hr^{-1}$
t½a	75 minutes	1.25 hr, so $k_a = \frac{0.693}{1.25} = 0.5544\ hr^{-1}$

THE WORKING, ONE LINE AT A TIME

UNIT

$$k_a = \frac{0.693}{1.25\ hr} = 0.5544\ hr^{-1};\ k = \frac{0.693}{7.3\ hr} = 0.0949\ hr^{-1};\ V_D = (3.2\ L/kg)(72\ kg) = 230.4\ L$$

The absorption half-life of 75 minutes is 1.25 hours; each rate constant is 0.693 over its half-life; the volume per kilogram is multiplied by the weight.

ALGEBRA

$$t_{max} = \frac{\ln(0.5544/0.0949)}{(0.5544 - 0.0949)\ hr^{-1}} = 3.8413\ hr$$

The peak has to be located before it can be evaluated; the time to peak depends only on k_a and k, so it is the same 3.84 hr for any dose of this drug, whatever the dose or the volume.

ALGEBRA

$$C_{max} = \frac{(0.85)(300\ mg)(0.5544\ hr^{-1})}{(230.4\ L)(0.5544 - 0.0949)\ hr^{-1}}\ (e^{-(3.84)(0.0949)} - e^{-(3.84)(0.5544)}) = (1.3354\ mg/L)(0.5756) = 0.7687\ mg/L$$

Reciprocal hours cancel between k_a and $(k_a - k)$, and mg over L leaves mg/L; the coefficient is 1.3354 mg/L, and the large volume, 230.4 L, keeps the first-dose peak below 1 mg/L.

> **DOES THE NUMBER MAKE SENSE?**
> The absorbed dose, 0.85 × 300 mg, in 230.4 L would be a little over 1 mg/L if it were all in the body at once; the peak must lie below that, and 0.7687 mg/L does. A value in the tens means the volume was not multiplied by 72 kg.

2b Her key

Answer: 0.6092 mg/L

WHAT THE STEM GIVES

weight	72 kg	$V_D = 3.2 \times 72 = 230.4$ L
D_0	300 mg	the dose in the numerator
τ	every 8 hours	t = τ = 8 hr in both exponents
duration	2 weeks	*not needed: long enough for steady state*
F	85% bioavailable	F = 0.85 in the numerator
V_D	3.2 L/kg	times 72 kg gives 230.4 L, the denominator
t½	7.3 hours	gives $k = \frac{0.693}{7.3} = 0.0949\ hr^{-1}$
t½a	75 minutes	1.25 hr, so $k_a = \frac{0.693}{1.25} = 0.5544\ hr^{-1}$

THE WORKING, ONE LINE AT A TIME

SETUP

$$C_{min} = \frac{F\,k_a\,D_0}{V_D(k_a - k)}\,(e^{-k\tau} - e^{-k_a\,\tau}),\ \text{with } \tau = 8\ \text{hr}$$

The trough of the first dose is the oral equation at the end of the 8-hour interval, just before the second dose.

ALGEBRA

$$C_{min} = (1.3354\ \text{mg/L})(e^{-(8)(0.0949)} - e^{-(8)(0.5544)}) = (1.3354\ \text{mg/L})(0.4680 - 0.0119) = 0.6092\ \text{mg/L}$$

The coefficient is the one from part a; after 8 hours the absorption term is almost gone and the elimination term, 0.4680, sets the trough.

> **DOES THE NUMBER MAKE SENSE?**
> Eight hours is about one elimination half-life of 7.3 hours and absorption is nearly complete, so the trough is a little under half the coefficient 1.3354 mg/L and below the first-dose peak; 0.6092 mg/L fits.

2c Her key

Answer: 1.64 mg/L

WHAT THE STEM GIVES

weight	72 kg	$V_D = 3.2 \times 72 = 230.4$ L
D_0	300 mg	the dose in the numerator
τ	every 8 hours	in t_{max}^{∞} and in the accumulation factor $\frac{1}{1 - e^{-k\tau}} = 1.880$
duration	2 weeks	*not needed: long enough for steady state*
F	85% bioavailable	F = 0.85 in the numerator
V_D	3.2 L/kg	times 72 kg gives 230.4 L, the denominator
t½	7.3 hours	gives $k = \frac{0.693}{7.3} = 0.0949\ \text{hr}^{-1}$
t½a	75 minutes	1.25 hr, so $k_a = \frac{0.693}{1.25} = 0.5544\ \text{hr}^{-1}$

THE WORKING, ONE LINE AT A TIME

SETUP

$$t_{max}^{\infty} = \frac{1}{k_a - k}\,\ln\left[\frac{k_a(1 - e^{-k\tau})}{k(1 - e^{-k_a\,\tau})}\right]$$

The steady-state peak is evaluated at the steady-state time to peak, which the interval enters through the two bracketed terms.

ALGEBRA

$$t_{max}^{\infty} = \frac{1}{0.4595\ hr^{-1}} \ln\left[\frac{(0.5544)(1 - e^{-(0.0949)(8)})}{(0.0949)(1 - e^{-(0.5544)(8)})}\right] = (2.176\ hr)(\ln 3.145) = 2.49\ hr$$

The ratio inside the logarithm, 3.145, is smaller than k_a over k, 5.84, so the steady-state peak comes earlier than the first-dose peak at 3.84 hr.

SETUP

$$C_{max}^{\infty} = \frac{FD_0}{V_D} \times \frac{1}{1 - e^{-k\tau}} \times e^{-k\, t_{max}^{\infty}}$$

The absorbed dose over the volume, raised by the accumulation factor and declined for the time to the steady-state peak.

ALGEBRA

$$C_{max}^{\infty} = \frac{(0.85)(300\ mg)}{230.4\ L} \times \frac{1}{1 - e^{-(0.0949)(8)}} \times e^{-(0.0949)(2.49)} = (1.107\ mg/L)(1.880)(0.7895) = 1.64\ mg/L$$

mg over L is mg/L; the accumulation factor 1.880 and the decline 0.7895 are pure numbers. The steady-state peak is above the first-dose peak of 0.7687 mg/L.

> **DOES THE NUMBER MAKE SENSE?**
> 0.85 × 300 mg in 230.4 L is 1.107 mg/L; the accumulation factor 1.880 nearly doubles it and the 2.49 hour decline 0.7895 takes some back, so a steady-state peak near 1.6 mg/L fits, above the first-dose peak.

2d

Her key

Answer: 1.17 mg/L

WHAT THE STEM GIVES

weight	72 kg	$V_D = 3.2 \times 72 = 230.4$ L
D_0	300 mg	the dose in the numerator
τ	every 8 hours	the accumulation factor 1.880 and the decline $e^{-k\tau}$ over 8 hr
duration	2 weeks	*not needed: long enough for steady state*
F	85% bioavailable	F = 0.85 in the numerator
V_D	3.2 L/kg	times 72 kg gives 230.4 L, the denominator
t½	7.3 hours	gives $k = \frac{0.693}{7.3} = 0.0949\ hr^{-1}$
t½a	75 minutes	1.25 hr, so $k_a = \frac{0.693}{1.25} = 0.5544\ hr^{-1}$

THE WORKING, ONE LINE AT A TIME

SETUP

$$C_{min}^{\infty} = \frac{F\,k_a\,D_0}{V_D(k_a - k)} \times \frac{1}{1 - e^{-k\tau}} \times e^{-k\tau}$$

The steady-state trough is the first-dose coefficient raised by the accumulation factor and declined for one full interval.

ALGEBRA

$$C_{min}^{\infty} = \frac{(0.85)(300\text{ mg})(0.5544\text{ hr}^{-1})}{(230.4\text{ L})(0.5544 - 0.0949)\text{ hr}^{-1}} \times \frac{1}{1 - e^{-(0.0949)(8)}} \times e^{-(0.0949)(8)}$$

$$= (1.3354\text{ mg/L})(1.880)(0.4680) = 1.17\text{ mg/L}$$

The coefficient, 1.3354 mg/L, is the one from part a; times the accumulation factor and the 8-hour decline it gives the trough, above the first-dose trough of 0.6092 mg/L.

DOES THE NUMBER MAKE SENSE?

The steady-state trough is the first-dose trough times the accumulation factor 1.880, so nearly twice the part b value, near 1.2 mg/L; 1.17 fits and lies below the steady-state peak.

2e

Her key

Answer: 1.46 mg/L

WHAT THE STEM GIVES

weight	72 kg	$V_D = 3.2 \times 72 = 230.4$ L
D_0	300 mg	the dose in the numerator
τ	every 8 hours	multiplies V_D and k in the denominator
duration	2 weeks	*not needed: long enough for steady state*
F	85% bioavailable	F = 0.85 in the numerator
V_D	3.2 L/kg	times 72 kg gives 230.4 L, the denominator
t½	7.3 hours	gives $k = \frac{0.693}{7.3} = 0.0949\text{ hr}^{-1}$
t½a	75 minutes	*not needed: the average does not depend on k_a*

THE WORKING, ONE LINE AT A TIME

SETUP

$$C_{avg}^{\infty} = \frac{FD_0}{V_D\,k\,\tau}$$

The average over one interval at steady state is the absorbed dose per interval over the volume of plasma cleared per interval; it needs no t_{max} and no accumulation factor.

ALGEBRA

$$C_{avg}^{\infty} = \frac{(0.85)(300\text{ mg})}{(230.4\text{ L})(0.0949\text{ hr}^{-1})(8\text{ hr})} = \frac{255\text{ mg}}{174.9\text{ L}} = 1.46\text{ mg/L}$$

L × hr^{-1} × hr leaves litres, so mg over L is mg/L; the average lies between the trough, 1.17, and the peak, 1.64 mg/L.

DOES THE NUMBER MAKE SENSE?

> 255 mg is absorbed every 8 hours and 174.9 L of plasma is cleared in that time, so the average is a little under 1.5 mg/L; it lies between the steady-state trough and peak.
